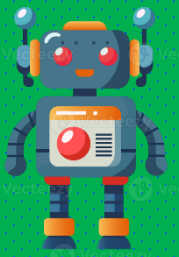


Smart Picks: Your AI-Powered Shopping Buddy



On Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

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Project Description

Project Title

Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

Company: Narosu Co. Ltd.

Project Description

- The e-commerce market is becoming increasingly competitive, requiring advanced recommendation systems.
- Large marketplaces use sophisticated AI models, but independent online store operators lack such solutions.
- This project aims to develop an **AI-powered product recommendation system** tailored for self-managed online stores.
- By analyzing customer preferences, the system will suggest products that align with their interests.

Project Objective

- Develop an AI-based search and recommendation system that analyzes user input sentences and recommends the **TOP 5** most relevant products based on semantic similarity.
- Utilize Natural Language Processing (NLP) and vector search technology to provide a highly refined search experience.
- Deploy in an on-premise environment to ensure data privacy and full control over the recommendation process.

Selected Technology Stack

Selected Model and Vector Store:

Component	Technology Used	Description
LLM (Large Language Model)	LLaMA 3.1 8B instant	Performs reasoning and generates responses based on user input and context
LangChain (LLM Framework)	LangChain	Connects LLMs to external tools and vector databases for enhanced reasoning and document access
Embedding Model	OpenAI text-embedding-ada-002	Converts text into dense vector representations using semantic similarity
Vector DB / Search	Qdrant	High-dimensional vector indexing with cosine similarity, metadata filtering, and hybrid search
Vector Search Library	Qdrant (built-in)	Optimized internal search within Qdrant; no separate library like FAISS needed
RAG (Retrieval-Augmented Generation)	LangChain + Qdrant + LLM	Retrieves semantically relevant documents from Qdrant to support LLM-generated answers
API Architecture	Gradio + FastAPI	Gradio handles UI and function binding; FastAPI supports additional backend API endpoints
GUI (Graphical User Interface)	Gradio	Provides interactive chatbot interface with support for image display and recommendation output

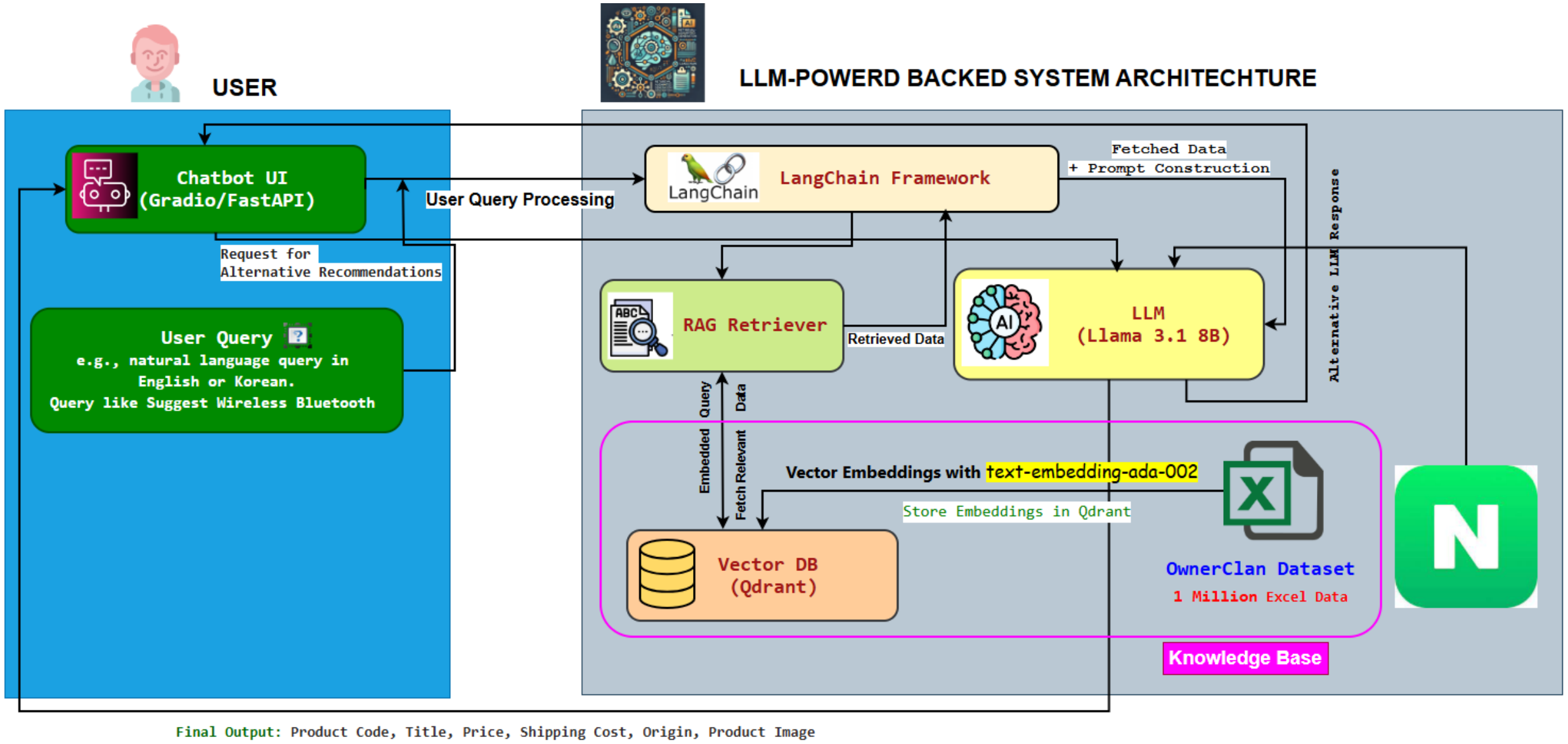
Requirements Fulfillment Summary - ✓ 100% Fulfilled

- Company Requirement Alignment
- LLM Setup (LLaMA 3.1 8B Instant)
- LangChain Integration
- OpenAI Embedding + Vector DB (Qdrant)
- Vector Search & RAG Implementation
- API Development (FastAPI + Gradio)
- Chatbot GUI with Image Display
- Recommendation Logic (OwnerClan 3 + Naver 2)
- Fallback logic for alternative recommendations
- Response Time Optimization (≤ 5s)
- Flexible model/embedding model architecture
- Mutiple languages support
- Modifiable attribute fields (title, code, origin, etc.)

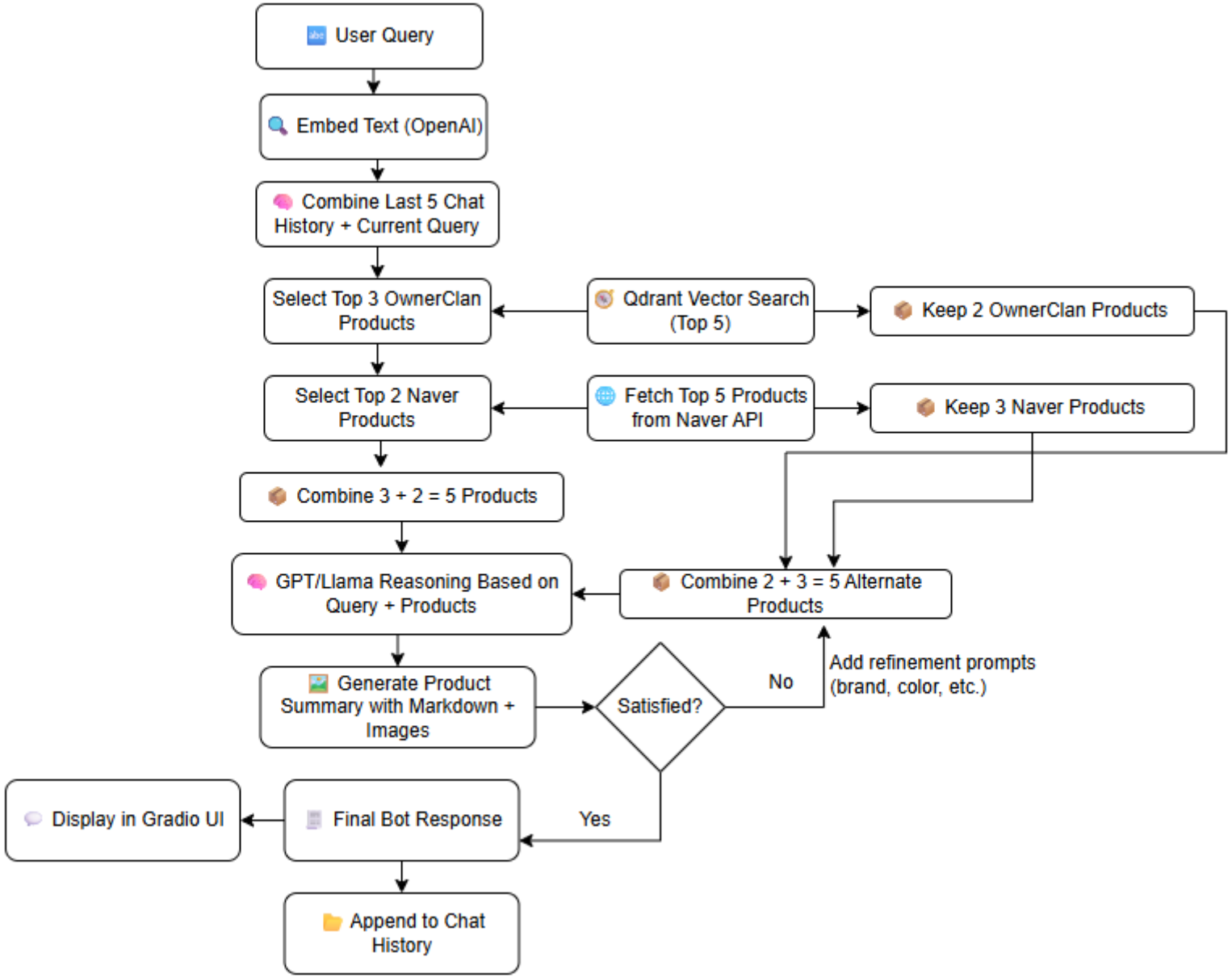
Extra Features:

1. Login/ Register Option
2. Mutiple Users Management
3. Chat History Option
4. Voice Input
5. Dedicated Naver Product Search Button
6. Evaluation Dashboard

System Configuration (AI Model) Diagram



Chatbot Working Pipeline



Evaluation Metrics in Product Recommendation System

Metric	Description	Use Case	Goal
Precision@K	% of top-K results that are relevant.	Measures how accurate the top product picks are.	High precision = fewer irrelevant suggestions.
Recall@K	Fraction of relevant products that appear in the top-K recommendations.	Measures how completely relevant products are captured.	High recall = fewer missed good products.
NDCG@K	Ranking quality considering position weights	Ensures the most suitable products appear first in the list.	Good rank order = better UX.
MRR@K	Evaluates how quickly the first relevant item appears in the ranking.	Ideal for single-answer queries like FAQs or product lookups	Higher MRR = relevant results appear earlier, improving user satisfaction.
MAP	Average of precision values at ranks of all relevant items (overall score). Mean of average precision across queries	Summarizes precision across positions — helpful for comparing system performance.	Balanced overview of recommendation quality.

Live Demo

Thank You for Listening.